AGRICULTURAL DRAINAGE · NOTE DR01

Need for Drainage — Waterlogging, Salinity and the Indian Situation

Course topics: CAE1.3 L14, IDE-502 T1, IDE-502 T3, IDE-502 T4, IDE-501 T7, IDE-515 T12, MDC-IV L16

What drainage is

Agricultural drainage removes excess surface water and groundwater from farmland. Drainage and irrigation are complementary: together they create a favourable soil-moisture regime, keep the salt balance in the root zone, give optimum growth of crops, plants and trees, and avoid harm to the environment.

Why crops need drainage

- Roots need a favourable environment to take up water and nutrients. Excess water or salt in the root zone stops roots working properly: growth and yield fall, seeds may fail to germinate and standing crops may wilt permanently.
- Waterlogging harms crops mainly through **poor aeration**. Normal growth needs an oxygen diffusion rate above 4×10^{-7} g/cm²/min; waterlogging blocks the connected air passages that allow it.
- **Salts:** poor-quality irrigation water adds salts, which change the soil's physical and chemical properties and cut water and nutrient uptake. Most plants grow less once the soil-water salt concentration exceeds about 5000 ppm.
- **The mechanism:** continual over-irrigation raises the water table; once it is within capillary-rise height, water moves up to the surface and evaporates, leaving salts in the surface and root zone — salinity or alkalinity, waterlogging and land deterioration. Adequate drainage is the remedy.
- A drainage system removes excess water and salts from the root zone and is essential for managing land and water and sustaining irrigated farming.

Type of drainage	How water is removed
Surface drainage	From the land surface
Subsurface (internal) drainage	By downward flow through the soil — horizontal (through tiles) or vertical (by pumping)

Drains remove only free (gravitational) water, which plants cannot use; they do not remove capillary water, so there is no danger of over-drainage even with closely spaced drains.

Waterlogging

- Land is waterlogged when the subsoil water table is at or near the surface and yields fall below normal — when infiltrating or seeping water fills the pores and drainage is inadequate or absent.
- **FAO/UNESCO (1973):** soils temporarily saturated, or with a permanent shallow water table.
- **National Commission on Agriculture (1976):** the water table rises enough to saturate the root-zone pores, restricting air circulation, lowering oxygen and raising carbon dioxide.

Government of India norms (1991)	Depth to water table
Waterlogged area	Within 2 m of the land surface
Potential waterlogging area	2–3 m
Safe area	Below 3 m

- Water-table depth can be expressed as pre-monsoon (April–May) depth, post-monsoon (Oct.–Nov.) depth, average seasonal or annual depth, or the sum of excess-water days above a set depth. The pre- and post-monsoon depths matter most.
- **Suggested season-linked norms:** waterlogged if ≤ 2 m pre-monsoon or ≤ 1 m post-monsoon; critical if 2–3 m pre-monsoon or 1–2 m post-monsoon; potential if 3–5 m pre-monsoon (unless water management is efficient).

How much it hurts

- A shallow water table is not always harmful in humid regions without salinity — capillary rise can supply part of the crop's ET, depending on texture and porosity.
- Deep-rooted crops (pulses, oilseeds) suffer when the water table comes within about 1 m. The permissible depth depends on soil texture and structure, climate, groundwater quality and the crop.
- Sensitive crops such as pigeon pea and maize lose 30–50% of yield when waterlogging lasts 24–48 hours.

Adverse effects

- Oxygen falls and CO_2 rises in the root zone; anaerobic conditions let harmful microbes dominate; physical, chemical and biological activity is disturbed, encouraging pests and diseases.
- Field operations become difficult or impossible.
- Capillary rise brings salts up from lower horizons or saline groundwater; secondary salinisation restricts moisture and nutrient uptake and causes toxicity.
- Alkali soils lose structure, infiltration becomes very low, so runoff increases and floods neighbouring crops.
- Yields fall or crops fail; fodder from salt-affected soils may carry high molybdenum and selenium and low zinc, causing livestock disease; the soil–water–crop relationship and the ecological balance are upset.

Causes of waterlogging and salinity

Excess rain, over-irrigation, seepage from reservoirs, canals and ditches, and inflow from adjoining land — shaped by topography, climate and human action. Chemical degradation is a greater risk in semi-arid and arid regions than in humid ones.

Cause	How it acts
(A) Topography and subsoil	Flat or concave land with restricted outflow; unfavourable geology, hard pans, low hydraulic conductivity; high natural soil salts
(B) Unprotected catchments	Deforestation, faulty cultivation and no conservation → erosion and silting of channels, ponds and reservoirs → waterlogging in flat land downstream
(C) Rainfall	High-intensity, short-duration monsoon storms in Asia → runoff collects and ponds
(D) Canal seepage	Unlined canals and seepage from higher irrigated land wet the low land beside them
(E) Man-made obstructions	Blocked drains, road and rail embankments, buildings and undersized culverts
(F) Injurious use of irrigation water	Field-to-field irrigation, over-application, untimely releases, ungated or unauthorised outlets, poor rostering, ignoring soil suitability
(G) Poor on-farm management	Bad land grading, irrigation–drainage layout and cropping patterns
(H) Drainage neglected in projects	Field, link and main drains not built or maintained → choked drains, cross bunds
(I) No integrated water management	Over-reliance on canal water instead of conjunctive use
(J) Wasted night canal flows	Unused night supplies waterlog land downstream (rare where warabandi is practised)
(K) No people's participation	Farmers not involved in maintaining field, link and main drains

Saline and alkali soils

Salts come from weathering of rocks and minerals, irrigation without drainage, salty canal water or poor groundwater, fossil salts, seepage from higher saline areas, coastal sea intrusion and fertilisers. Many low-lying areas with high potential have become unproductive and need drainage plus soil management.

Soil	pH	EC (dS/m)	ESP	Main salts
Saline	< 8.5	> 4	< 15	Chlorides and sulphates of Na, Ca, Mg
Alkali (sodic)	> 8.5	< 4	> 15	Bicarbonates, carbonates and silicates of sodium
Saline-alkali	Variable	> 4	> 15	Both

The notes give ESP "≈ 15" for saline-alkali soils; the standard USSL limit is ESP > 15 (with EC > 4 dS/m).

- Salinity depends on total salt (salinity), sodium carbonate and bicarbonate relative to Ca and Mg (sodicity), and toxic ions such as chloride and boron; it is expressed as electrical conductivity (EC) or as ppm or mg/L.
- Secondary salinisation comes from poor water, inadequate leaching, canal seepage, a shallow water table and high evaporation — in irrigated semi-arid and arid areas the water table rises and evaporation leaves salts in the root zone.
- Sodium is the most hazardous soluble constituent; its effect depends on its ratio to Ca^{2+} and Mg^{2+}, measured by the **Sodium Adsorption Ratio (SAR)**.

The problem in India

- **National Commission on Agriculture (1976):** about 6.0 million ha waterlogged — 3.4 million ha from surface-water stagnation and 2.6 million ha from a rising water table.
- **1991 estimate:** 2.46 million ha affected in irrigation command areas.
- Concentrations: the Chambal command (Rajasthan and Madhya Pradesh), where waterlogging grew after canal irrigation began (1968); coastal sea-water intrusion; poorly drained low-lying coastal rice belts.

Anti-waterlogging work in India

- 1925: Punjab enquiry committee on waterlogging; field experiments at Chakawali and a laboratory at Lahore. After Independence: the Land Reclamation, Irrigation & Power Research Institute, Amritsar.
- Schemes with International Development Agency support (before 1964): about 3,220 km of drains, embankments and flood works, surface drains, subsurface pumping stations, canal lining, tubewells and seepage drains, benefiting ≈ 1.6 lakh ha.
- **Punjab** (since 1960): 4,480 km of surface drains by 1971 helped check the rising water table, and farmers installed shallow tubewells. **Haryana — Jagadhri tube-well scheme:** pumped waterlogged areas to supplement the Western Yamuna Canal and encouraged private tubewells.
- **Gujarat:** the Mahi canal project (Cambay–Mater Bhal) needed leaching and drainage before irrigation expanded; the Shetrunji canal project (Saurashtra) drained about 7,000 ha of shallow water table with about 22 km of open drains 1.2–1.8 m deep. **Kerala — Kuttanad:** large propeller pumps drain monsoon waterlogging.
- States generally use flood embankments, drainage pumping and seepage drains along canals.
- **Policy:** drainage was integrated with irrigation planning only late in the Sixth Plan (1980–85); since the Eighth Plan (1992–97), adequate drainage is mandatory in project design and needs Central Government/CWC clearance, with MoEF approval for environmentally sensitive projects.

Research and pilot projects

- Tile-drainage pilots in Himachal Pradesh (Magwain, Majhiyal, Bangrota) of 25–50 ha each.
- Manjri scheme near Pune (1930): tile drains to open drains to the Mutha river — lowered the water table and salinity and raised production.
- Ranbirpura (Patiala) reclamation project (~800 ha) with surface and subsurface drainage; Punjab Agricultural University (1966) research on drainage criteria, deep open interceptor and relief drains.
- ICAR All-India Coordinated Research Project on Agricultural Drainage (late 1970s) — stations in representative problem regions generating field-ready information.

Reclaiming saline land

- **Coastal saline land** — affected by tidal sea water and toxic chlorides, often with high pH and exchangeable sodium — first needs embankments with one-way gates that keep the sea out but let monsoon water drain; many protected areas stay uncultivated for want of drainage and leaching.
- **Saline soils generally:** control the water table (preventive and remedial), leach salts to keep the root-zone balance, follow suitable agronomic practices, grow salt-tolerant crops, and monitor, maintain and correct after reclamation.

Salt tolerance	Crops
Tolerant	Barley, Bermuda grass, cotton, spinach, date palm, sugarbeet
Semi-tolerant	Wheat, oats, sorghum, maize, rice, tomato, cabbage, cauliflower, cucumber, peas, grapes, sunflower, Sudan grass, pomegranate
Sensitive	Apple, orange, lemon, pear, field beans, green beans, radish

Sources: ICAR eCourse 'Drainage Engineering' — Dr M. K. Jha (IIT Kharagpur), reviewed by Prof. K. Yellareddy (Director A&R, WALAMTARI, Hyderabad); AgriMoon e-book. USDA Handbook 60 (USSL) for soil salinity classes. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Under the 1991 GoI norms, an area is waterlogged when the water table is within

(a) 1 m (b) 2 m
(c) 3 m (d) 5 m

Q2. Normal crop growth needs an oxygen diffusion rate above about

(a) 4×10^{-7} g/cm²/min (b) 4×10^{-3} g/cm²/min
(c) 4 g/cm²/min (d) 0.4 g/cm²/min

Q3. Saline soils have

(a) pH > 8.5, ESP > 15 (b) EC > 4 dS/m, ESP < 15, pH < 8.5
(c) EC < 4, ESP > 15 (d) EC < 1, pH < 7

Q4. The NCA (1976) put India's waterlogged area at about

(a) 0.6 Mha (b) 2.46 Mha
(c) 6.0 Mha (d) 16 Mha

Q5. Which crop is salt-sensitive?

(a) Barley (b) Date palm
(c) Sugarbeet (d) Field beans

Q6. Drainage became mandatory in irrigation project design from the

(a) Third Plan (b) Sixth Plan
(c) Eighth Plan (d) Twelfth Plan

Answers

Q1 — (b) 2 m
Q2 — (a) 4×10^{-7} g/cm²/min
Q3 — (b) EC > 4 dS/m, ESP < 15, pH < 8.5
Q4 — (c) 6.0 Mha — *3.4 Mha surface stagnation + 2.6 Mha high water table.*
Q5 — (d) Field beans
Q6 — (c) Eighth Plan